Home Reading 09 — Optional Root Locus and Robustness

Preparation

Use this optional unit after Lecture 8. Allow about 25–30 minutes. It is separate from Project 2 and does not change Tutorial 1–8 submission credit.

Read selectively

Lecture 9: slides 5–6 and 8–12 for gain, poles and response; slides 18–19 for saturation; slides 23–25 for uncertain gain. Try the questions below before reading the worked calculations.

Textbook support: Ogata, Modern Control Engineering, 5th ed., printed pp. 270–271 (PDF pp. 281–282). Read the root-locus definition and characteristic equation. The angle and magnitude construction rules are further reading.

Check your understanding

1. What does a root locus show? Distinguish an open-loop pole from a closed-loop pole.

2. For poles $-1 \pm 3j$, what sets the decay rate and oscillation frequency? Would $-1 \pm 5j$ necessarily settle faster?

3. What is the difference between requested and delivered actuator input? Why does clipping change the response predicted by a linear transfer function?

4. What does robust feasibility mean? Why might the worst parameter value for tracking differ from the worst value for overshoot?

Prepare for Tutorial 9

Write a pole-location sketch and one prediction about increasing gain. Try Q1 and the first polynomial match in Q2; identify one step you cannot yet explain. Q3 applies the same method to uncertain damping. Extension A is further practice.

You are ready when you can explain the four checks without copying the slides and show your own first step for Q2. Bring any unresolved question to the discussion.

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November 2026